

Mission Scenarios

How to use this section

Each scenario is a step-by-step walkthrough that touches the full flow: Constellation/Formation → Targeting → Controller (PD/DRL) → Fault Injection → Detection & XAI → Reassignment → Safe Mode & Solar → Plots/Results/Export → Vizard. Follow the steps, take the suggested screenshots, and paste the “Insights” into your demo notes or report.

Pre-run Checklist

- Make sure Vizard is set up to display only assigned targets (global comm lines/cones OFF) and that targets created in the GUI have the correct assigned_to list.
- Make sure that the active mode is visible in the header/status area and that the DRL toggle (Rule-based ↔ DRL) functions.
- Make sure the XAI tab opens without any issues after loading the Keras Anomaly Detection model.
- Make sure that auto-exports end up in a clean directory, open Results/Export and select a run folder. For comparability, make a note of the time window you intend to run (e.g., 120–180 s).

Scenarios 1: Nominal Imaging Mission (PD vs DRL A/B)

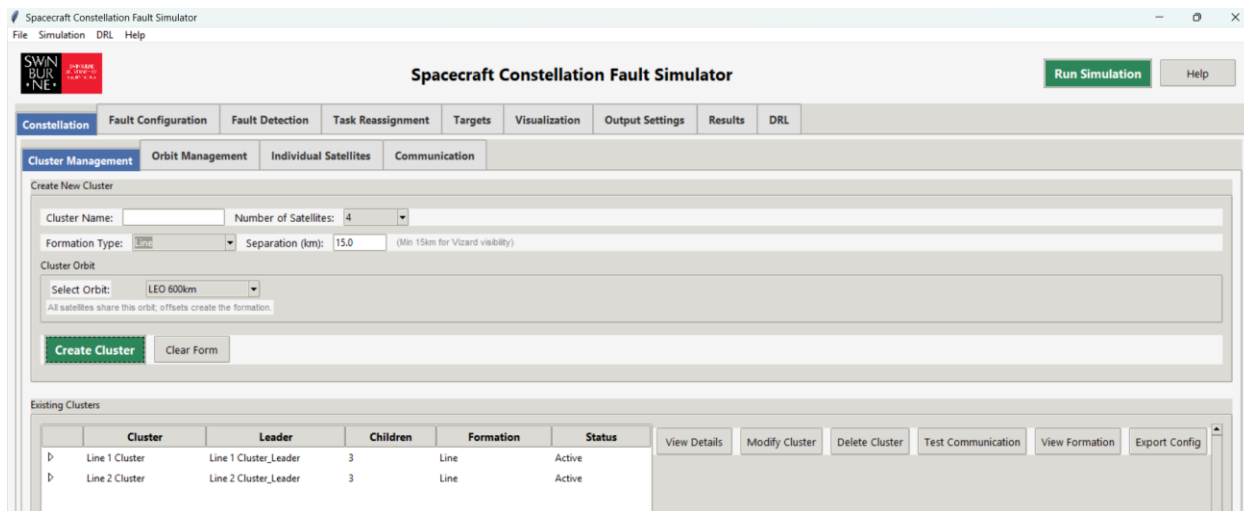
Goal

Show a clean, nominal run on the same targets using PD first, then DRL, and compare attitude settling and task throughput.

Steps

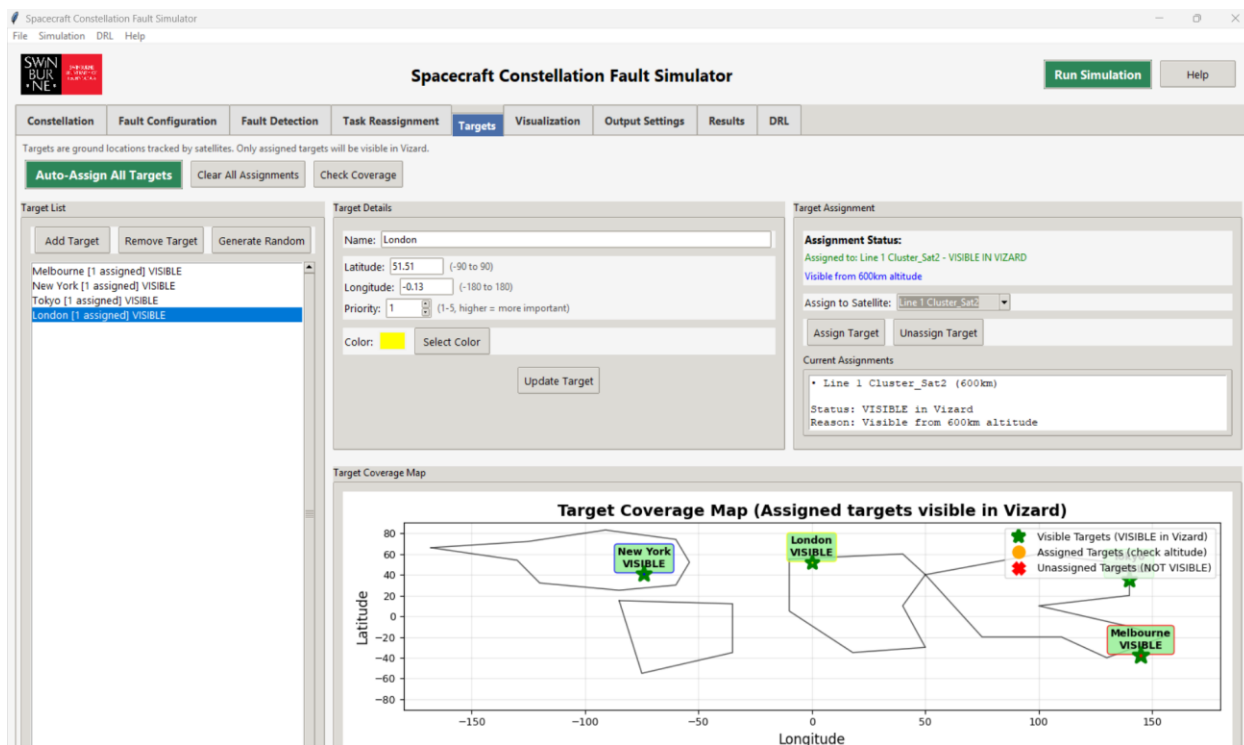
Constellation / Formation

Open Constellation → pick Line (or preferred) formation, 2 clusters (8 sats total), Select Orbit LEO 600km (or preferred). Verify cluster status messages appear (simulation start, formation applied).



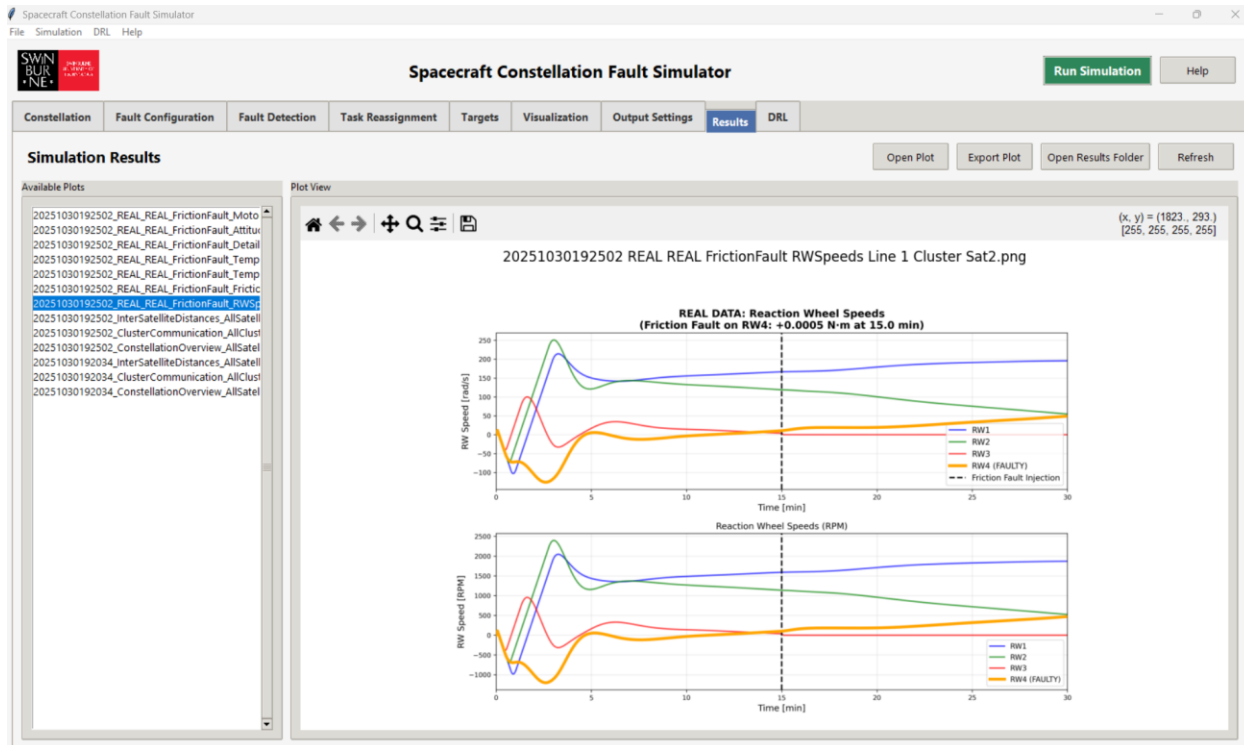
Targeting

Open Target → Assign two targets to Sat-1 and two to Sat-2 (leave Sat-3/4 idle as controls).
Confirm markers display and only assigned targets show as active for each satellite.



Plots & Export (Run A)

In Results/Plots, capture Attitude Error, RW torque/power, and Task Progress.



Controller toggle (DRL)

Reset sim. Toggle to DRL (PPO). Run the same scenario/time window (match the PD duration).

Plots & Export (Run B)

Capture the same plots and export as RunB_DRL.

(Optional) Open DRL tab to show live metrics (episode reward, policy entropy).

Vizard

Open the Vizard file: verify only assigned targets appear linked/active per satellite; take one still at mid-run.

Evidence to capture

Formation view at start, target markers, DRL toggle state; PD vs DRL Attitude Error (overlay or side-by-side), Task Progress, RW power; Vizard still showing assigned-only targeting.

Insights (paste into slides)

DRL vs PD: show time-to-stabilise attitude and throughput differences; the GUI supports clean A/B testing (toggle + reproducible exports); targeting visuals match assignment (no “auto-target everything” artefact).

Scenario 2 — Battery Fault → Detection → DRL Reassignment → Safe Mode & Solar Recovery

Goal

Demonstrate the end-to-end autonomy flow: a battery fault triggers detection; DRL reassigns tasks; the affected sat enters Safe Mode at low SoC, then recovers via solar.

Steps

Constellation / Formation

Use Diamond-4 (or any) with 4 satellites; confirm cluster status messages and initial FPS.

The screenshot displays the 'Spacecraft Constellation Fault Simulator' interface. The 'Cluster Management' tab is active, showing a 'Create New Cluster' form and a table of 'Existing Clusters'.

Create New Cluster Form:

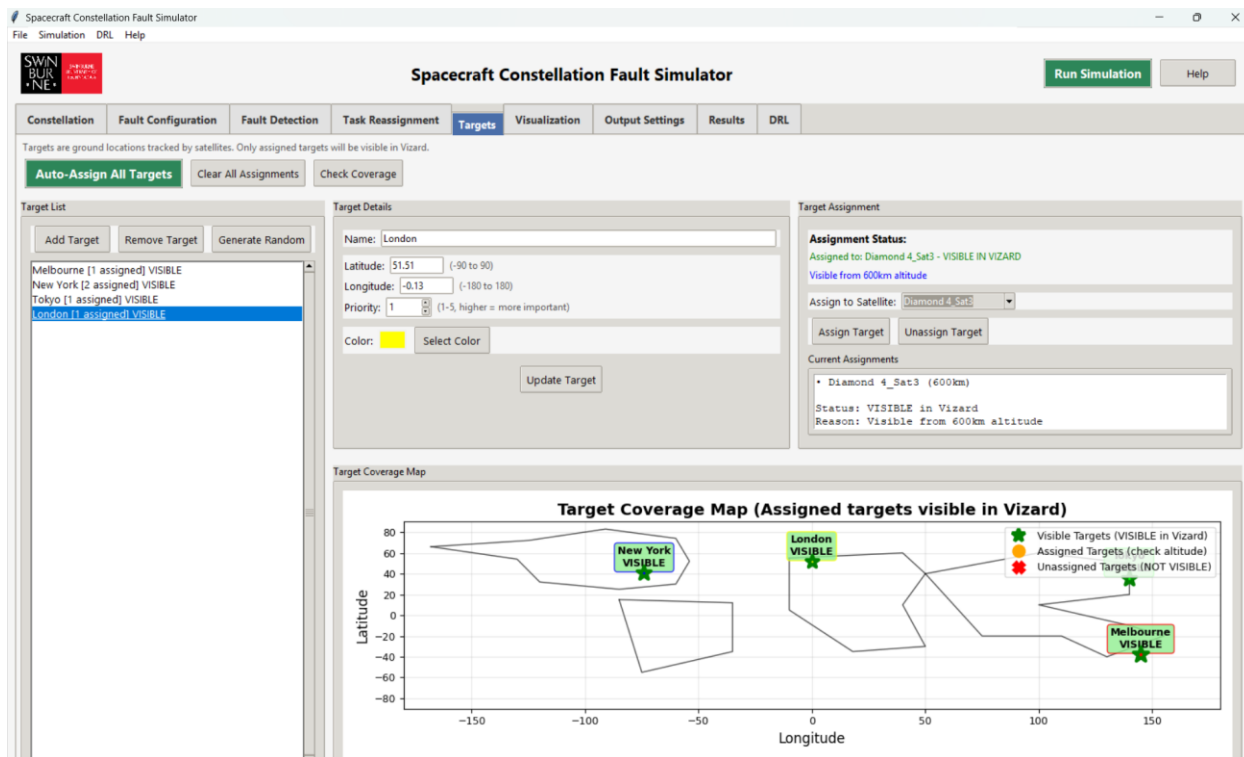
- Cluster Name:
- Number of Satellites:
- Formation Type:
- Separation (km): (Min 15km for Vizard visibility)
- Cluster Orbit:
- Buttons:

Existing Clusters Table:

Cluster	Leader	Children	Formation	Status	View Details	Modify Cluster	Delete Cluster	Test Communication	View Formation	Export Config
Diamond 4	Diamond 4_Leader	3	Diamond	Active						

Targeting

Assign 1–2 targets to each of Sat-1/2/3; keep Sat-4 lightly loaded to act as a reassignment sink.



Detection & XAI

Fault Detection → load Keras model; run a quick test inference (sanity check).

Open XAI tab and keep it ready (you'll snapshot it after detection).

Controller toggle

Set DRL (PPO) so reassignment decisions are agent-driven.

Inject Battery Fault

Faults → Battery Capacity → pick Sat-2 → degrade capacity over time (or a step drop to trigger sooner). Note the timestamp when injection starts.

Spacecraft Constellation Fault Simulator

File Simulation DRL Help

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Spacecraft Constellation Fault Simulator

Run Simulation Help

Constellation Fault Configuration Fault Detection Task Reassignment Targets Visualization Output Settings Results DRL

Fault Configuration Fault Summary Cluster Faults

Satellite Selection

Select Cluster: Diamond 4 Show All

Select Satellite: Diamond 4_Sat2

Cluster: Diamond 4 | Role: child | Orbit: LEO 600km

Fault: Battery on RW4

Fault Configuration

☒ Enable Fault

Fault Type: ☐ Friction ☐ Power Limit ☐ Encoder ☒ Battery

Fault Description

Battery fault simulates increased power drain. Default: 50 W additional drain.

Fault Parameters

Power Drain (W): 50.0

Fault Time (min): 15.0

Periodic Fault

☐ Enable Periodic Fault

Interval (sec): 360

Magnitude: 0.1

Wheel Number: 2

Apply Fault Configuration ? Apply to All Satellites Apply to Cluster

Spacecraft Constellation Fault Simulator

File Simulation DRL Help

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Spacecraft Constellation Fault Simulator

Run Simulation Help

Constellation Fault Configuration Fault Detection Task Reassignment Targets Visualization Output Settings Results DRL

Fault Configuration Fault Summary Cluster Faults

Fault Summary Overview

Refresh Summary Clear All Faults

Statistics

Total: 4 satellites | Faults: 1 enabled

Satellite Fault Status

Satellite	Cluster	Role	Fault Status	Type	Wheel	Time (min)	Magnitude
Diamond 4_Leader	Diamond 4	leader	Disabled	Friction	RW4	15.0 min	0.0005 N.m
Diamond 4_Sat2	Diamond 4	child	ENABLED	Battery	RW4	15.0 min	50.0 W
Diamond 4_Sat3	Diamond 4	child	Disabled	Friction	RW4	15.0 min	0.0005 N.m
Diamond 4_Sat4	Diamond 4	child	Disabled	Friction	RW4	15.0 min	0.0005 N.m

Observe

Detection flags Sat-2; open XAI to show LIME/SHAP (why/which features).

Reassignment: DRL shifts Sat-2 tasks to Sat-4 (or best candidate); confirm in the reassignment/cluster view.

Safe Mode: at $\text{SoC} \leq 20\%$, Sat-2 camera OFF; Solar Power panel shows charging; the sat later exits Safe Mode as SoC recovers.

Results & Export

Export Battery SoC, Camera state, Task Progress, Reassignment log, and Detection timeline to folder (Excel + images). Name folder RunC_BattFault.

Vizard

Confirm Sat-2 no longer shows active links to targets while in Safe Mode; other sats assume those targets; capture one still pre- and post-reassignment.

Evidence to capture

Detection screen (before/after), XAI panel for the event, who took over Sat-2's tasks, Solar/SoC plots during Safe Mode and recovery, Excel export with detection events + SoC trace.

Insights

End-to-end autonomy: Detect → Explain → Reassign → Survive → Recover. Safe Mode behaviour is visible and data-backed (SoC threshold, camera OFF/ON, solar charging curve). DRL reduces missed tasks and time-to-recover vs rule-based fallback.

Scenario 3 — Power-Limit/Friction Fault under Load (8–16 sats), DRL + Cluster Plots

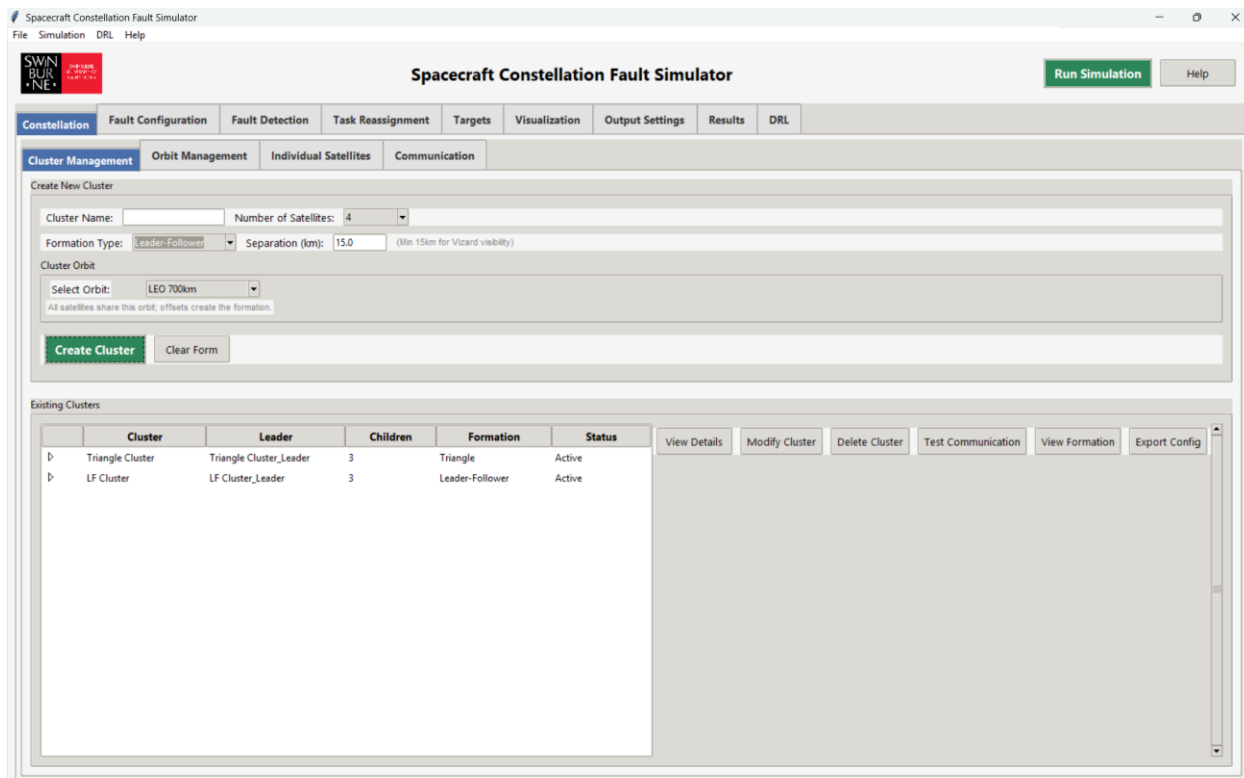
Goal

Show system resilience and performance at scale, with DRL handling faults and cluster plots providing situational awareness.

Steps

Constellation / Formation

Set 2 clusters × 4 satellites (8 total); choose a formation per cluster; verify stable starting FPS.



Targeting

Assign targets unevenly (Cluster-A heavier) to highlight reassignment when faults occur.

TARGET COVERAGE ANALYSIS REPORT

Target: Melbourne
 Location: -37.81°, 144.96°
 Assigned to: Triangle Cluster_Leader, Triangle Cluster_Sat2, Triangle Cluster_Sat3
 Visibility: VISIBLE in Vizard
 Coverage: Visible from 700km altitude

Target: New York
 Location: 40.71°, -74.00°
 Assigned to: Triangle Cluster_Leader, Triangle Cluster_Sat2, LF Cluster_Sat3
 Visibility: VISIBLE in Vizard
 Coverage: Visible from 700km altitude

Target: Tokyo
 Location: 35.68°, 139.77°
 Assigned to: Triangle Cluster_Leader, Triangle Cluster_Sat2, Triangle Cluster_Sat4, LF Cluster_Leader
 Visibility: VISIBLE in Vizard
 Coverage: Visible from 700km altitude

Target: London
 Location: 51.51°, -0.13°
 Assigned to: Triangle Cluster_Sat3, Triangle Cluster_Sat4, LF Cluster_Sat2
 Visibility: VISIBLE in Vizard
 Coverage: Visible from 700km altitude

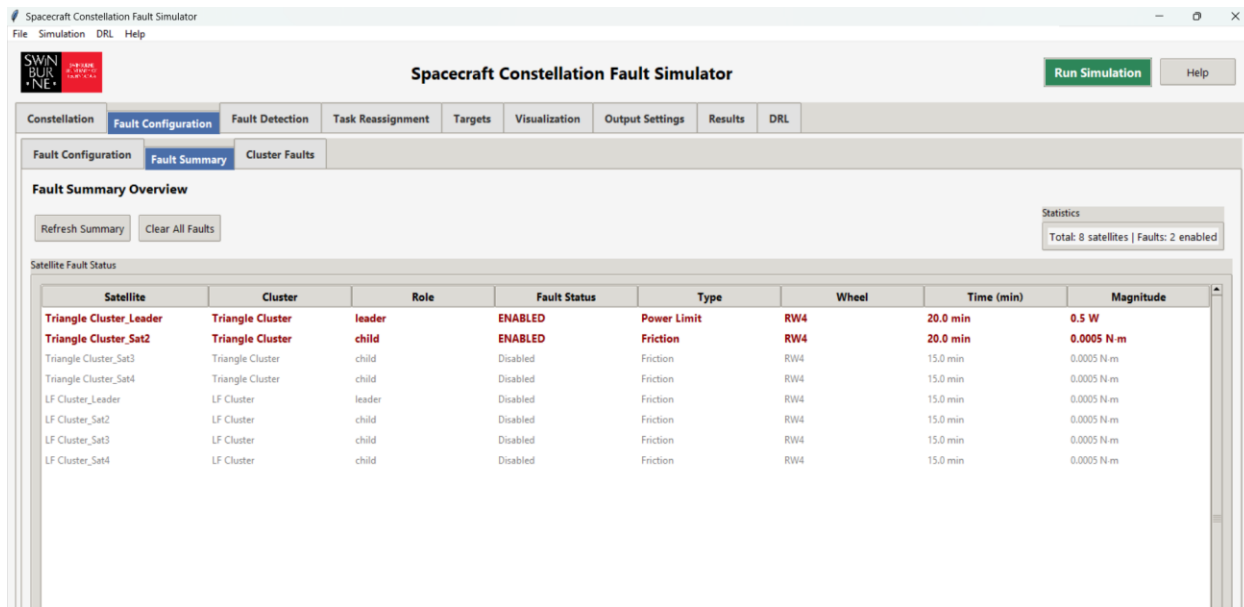
SUMMARY:
 Total Targets: 4
 Assigned Targets: 4
 Visible Targets: 4
 Unassigned Targets: 0

Controller & Tabs

Toggle DRL ON; open Cluster Plots, Detection, and DRL tabs.

Inject Faults

Apply Power-Limit to a leader in Cluster-A; add a Friction Stall to another sat in the same cluster (stagger timings by ~20–30 s).



Observe

Detection raises flags; DRL reassigns tasks within/ across clusters. Cluster Plots show load shift and recovery slopes; Task Progress remains stable (no large backlog growth).

Results & Export

Export Cluster Plots, Task Progress, Fault timelines, and Excel logs; save as RunD_LoadFaults.

Vizard

Check that only assigned targets are active per sat and the scene remains readable at scale; grab one still at peak load.

Evidence to capture

Cluster load plot before/after faults; Task Progress/Remaining-tasks heatmap; DRL tab metrics; detection/reassignment screens; export folder listing.

Insights

Under load, DRL maintains mission throughput via quick reassignment. Cluster-level visualization helps spot bottlenecks and confirm the effect of decisions. The system remains responsive (FPS/latency) at 8–16 sats with faults active.

Scenario 4 — Encoder Fault & Sensor Robustness (Detection/XAI stress)

Goal

Exercise Encoder corruption and verify the Detection/XAI stack isolates sensor-driven anomalies without false positives.

Steps

Constellation / Formation

Line-4, 4 sats; stable FPS.

Targeting

Assign one high-rate imaging target to Sat-1 (sensitive to pointing); light tasks for others.

Controller & Tabs

Start with PD for predictability.

Inject Encoder Fault

Faults → Encoder on Sat-1; introduce drift/noise.

Observe

Detection flags Sat-1; XAI shows feature importance skewed to encoder/attitude deltas.

Switch to DRL and rerun a short window to compare control recovery.

Results & Export

Export Attitude Error, Encoder signal, Detection/XAI snapshots, Task Progress → RunE_Encoder.

Vizard

Confirm Sat-1 struggles to hold target under PD; DRL mitigates drift better.

Insights

Detection pinpoints sensor anomaly; XAI explains why; DRL shows improved recovery vs PD for pointing-sensitive tasks.

Scenario 5 — Anomaly Models A/B (Keras vs Random Forest) + Threshold Tuning

Goal

Compare two detection models on the same events; tune thresholds to balance precision/recall and demonstrate XAI consistency.

Steps

Constellation / Formation

Reuse Scenario-2 layout (Battery event); keep assignments constant.

Model A (Keras)

Run Battery degradation; collect detections, XAI, false positives if any; export to RunF_Keras.

Model B (Random Forest)

Switch model; repeat the same run window; adjust threshold to match desired sensitivity; export to RunF_RF.

Compare

Place Detection timelines side-by-side; list differences; include XAI visuals.

Insights

Show operational trade-offs; record chosen threshold and why (fewer false alarms vs earlier catch), with XAI alignment across models.

Scenario 6 — Target Area Management & Assignment Audit (GUI ↔ Vizard)

Goal

Demonstrate precise assignment: only assigned targets are acted on and visualised; validate recent targeting fix.

Steps

Constellation / Formation

Two clusters, 4–6 sats.

Targeting

Create 6–8 targets; assign disjoint sets per satellite (document in notes).

Toggle an optional “max assignees” rule if available; keep proximity auto-assigners OFF.

Run & Observe

Confirm each sat only engages its assigned targets; watch GUI markers for stability.

Open Vizard to verify assigned-only links; no “auto-target as you pass by.”

Stress

Increase targets; confirm stability and performance.

Results & Export

Save assignment map, Task Progress, and a short note on FPS/latency → RunG_TargetAudit.

Insights

Assignment is authoritative (GUI → sim → Vizard); visuals and logs match; performance remains stable as target count grows.

Scenario 7 — Communications & Handshake Visualization (Situational Awareness)

Goal

Show LoS/handshake overlays and how communication states relate to cluster leadership and task flow.

Steps

Constellation / Formation

Two clusters; designate leaders per cluster if applicable.

Targeting

Assign targets unevenly (Cluster-A heavier) to highlight reassignment when faults occur.

Comms Visuals

Enable communication overlays; start the run; take snapshots of link state changes as sats manoeuvre.

Results & Export

Save comm overlays and a timeline note of link up/down events → RunH_Comms.

Insights

Operators can read network state at a glance; leadership and task routing adapt when links fluctuate.

Notes & gotchas (to keep scenarios clean)

- Target assignment: ensure every target has the correct assigned__{to} list; unassigned targets should not appear active.
- Vizard visuals: keep global comm lines/cones OFF so only assigned targets look active; your fix already handles this.
- Reproducibility: name runs consistently (e.g., RunA_PD, RunB_DRL, RunC_BatteryFault) so exports line up with slides and JIRA evidence.
- Performance: when testing 8–16 sats, jot down FPS and step latency so you can claim stability with numbers; note any thermal throttling or plotting slowdowns.

Feature coverage (at a glance)

- Formations/Constellation: Scenarios 1–3, 6–7
- Targeting (assignment-only): Scenarios 1, 6
- Controller PD/DRL toggle: Scenarios 1–3, 4
- Fault Injection (all four): Battery (2), Power-Limit + Friction (3), Encoder (4)

- Detection + multi-model + XAI: Scenarios 2, 4, 5
- DRL Reassignment: Scenarios 2–3
- Safe Mode + Solar: Scenario 2
- Cluster/Comm Visuals: Scenarios 3, 7
- Plots/Results/Export + Vizard: All scenarios